

BI CT configuration for A1-ESS-G2



1. GEN 2 BI CT configuration

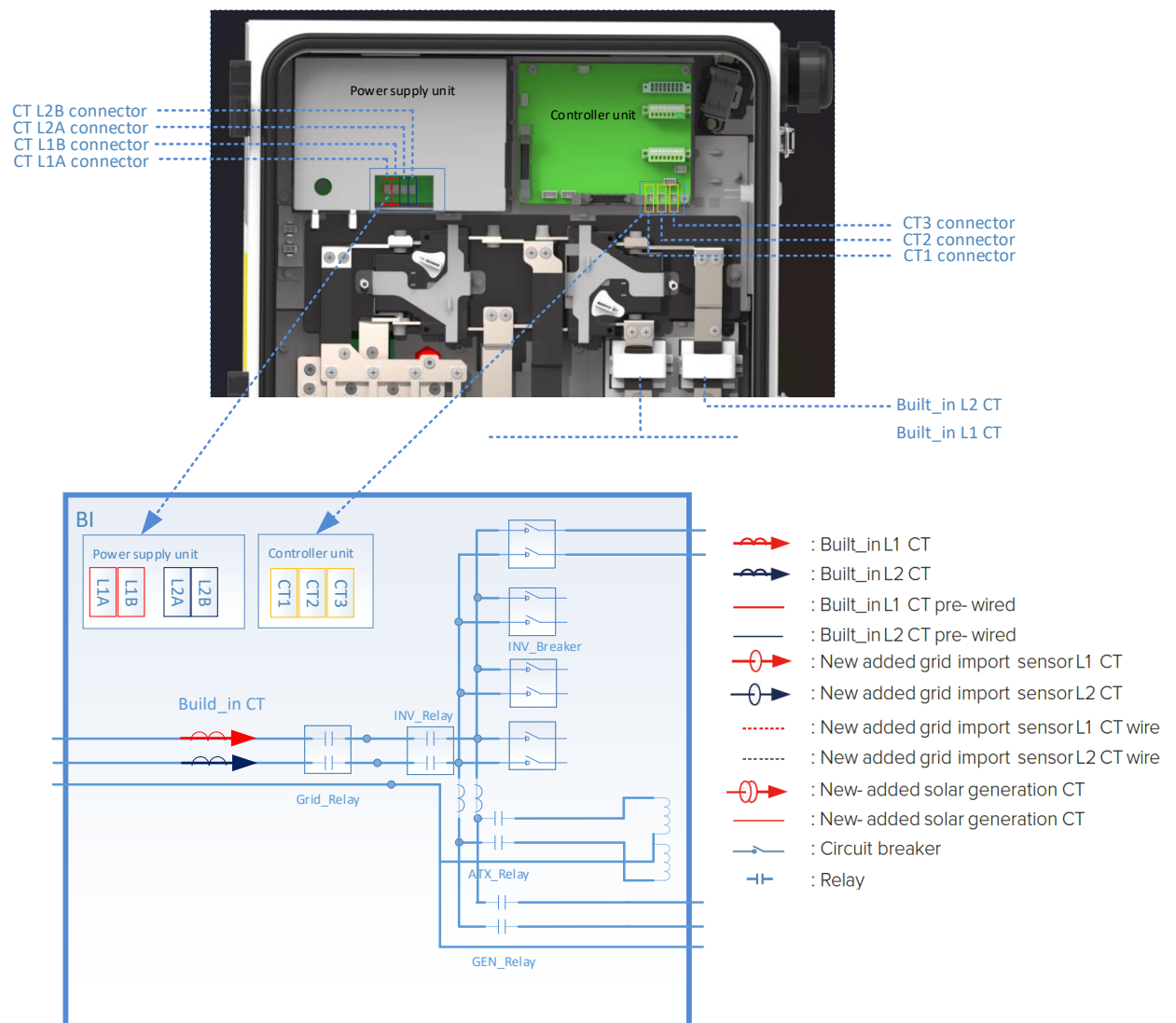
GEN 2 BI has built-in meter which measures the grid input power.

- 1. Whole home backup application, it is no need to configure the CT.

But for the follow 5 applications:

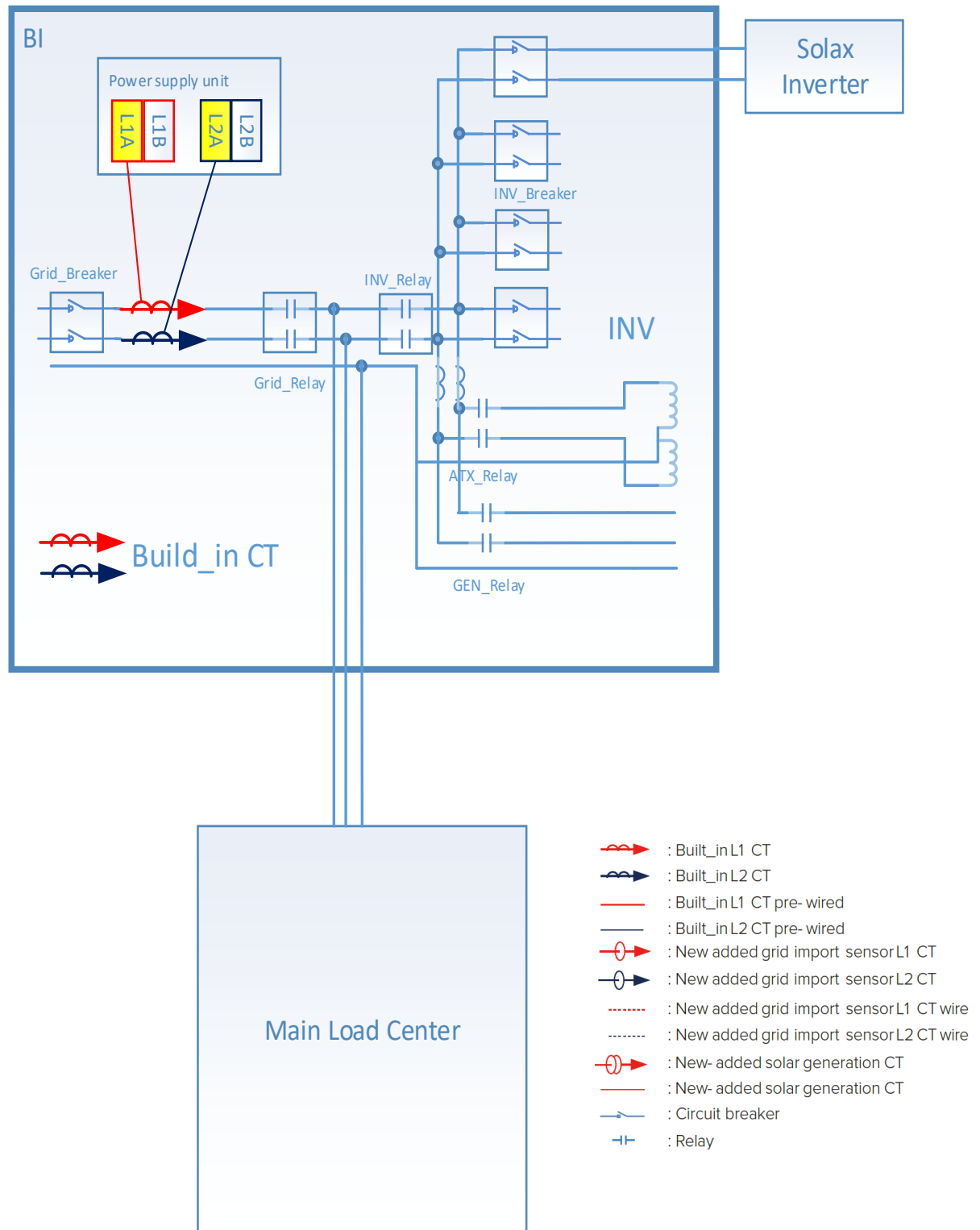
- 2. Whole home backup with third party grid tied solar inverter
- 3. Partial home backup
- 4. Partial home backup with third party grid tied solar inverter
- 5. Partial home back up with different panels
- 6. Partial home back up with third party grid tied solar inverter, with different panels

the built-in meter does not measure the import current of the whole home. As a result, the energy management strategy, like self-use and time of use, may not work. So a brief CT configuration is introduced to make sure the Generation 2 energy storage system works as desired.



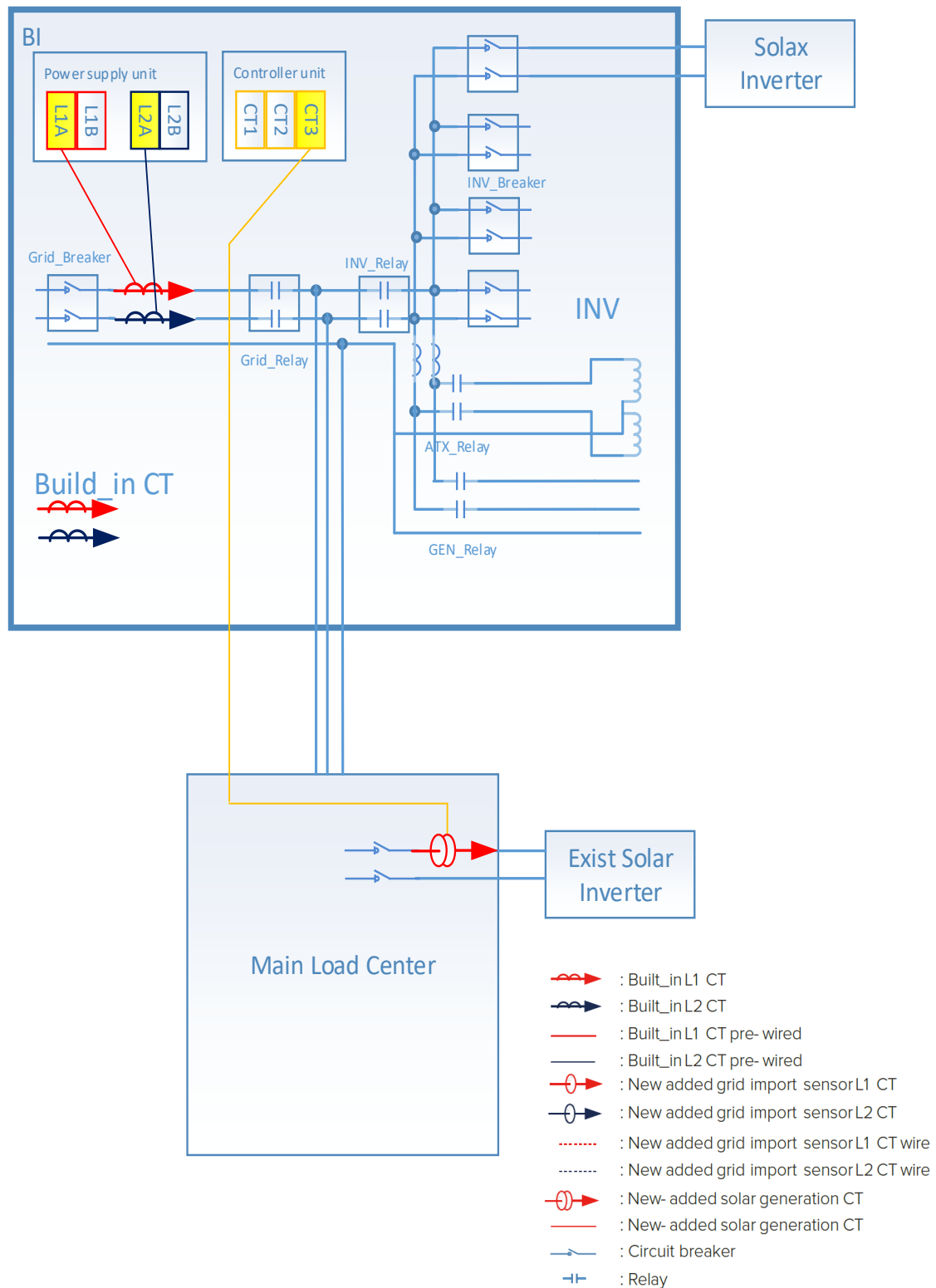
2. CT configurations

2.1 Whole home backup



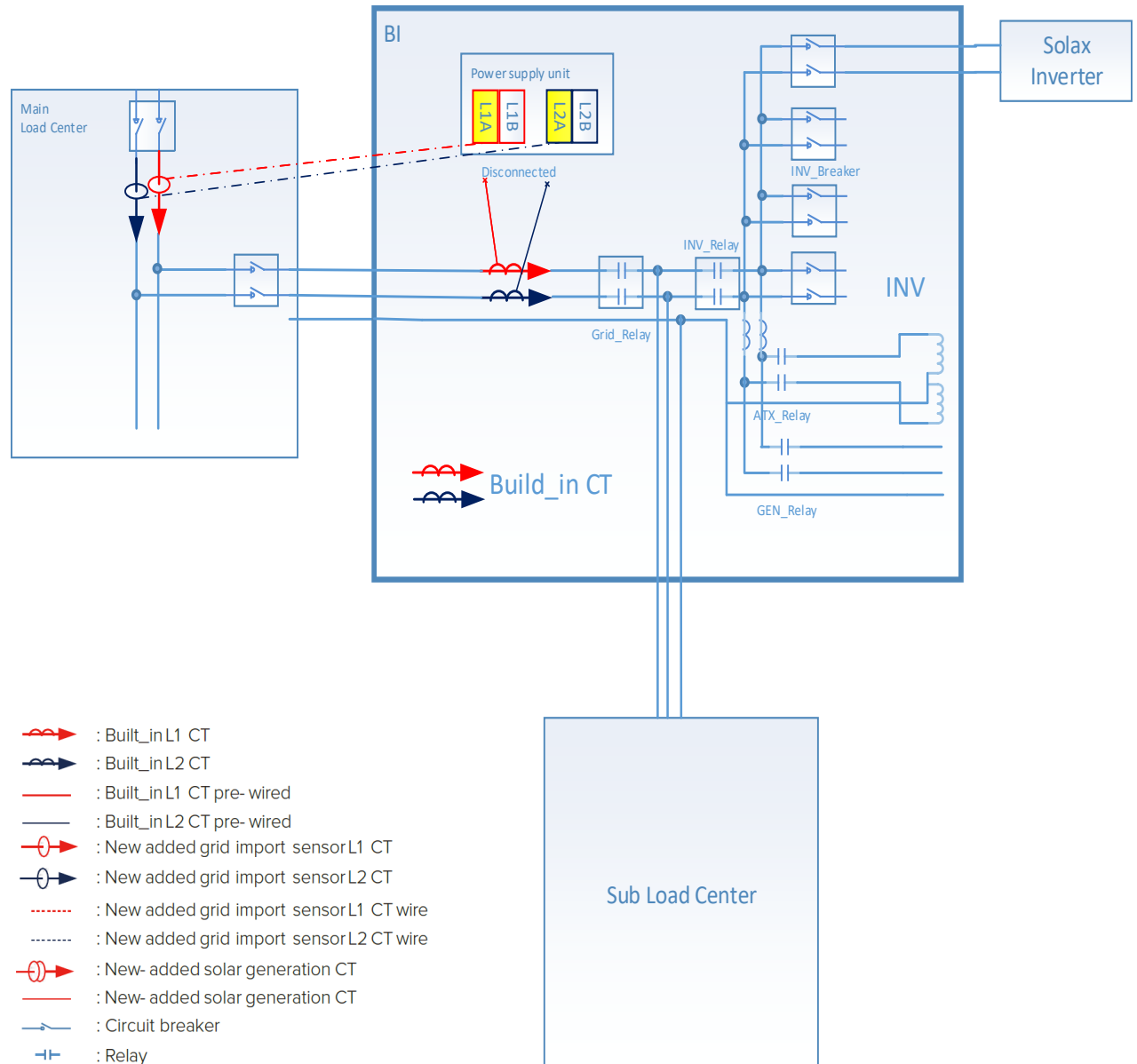
2.2 Whole home backup with third party grid tied solar inverter.

Need a new extra CT to monitor the current of solar inverter. This new extra CT should be connected to “CT3” connector on the controller unit. And the “CT3” clamp must connect to the grid tied solar inverter L1.



2.3 Partial home backup

Disconnect the pre-installed CT and re-connect the new added CT to the L1A and L2A respectively. Need two extra CT to monitor the grid input currents.

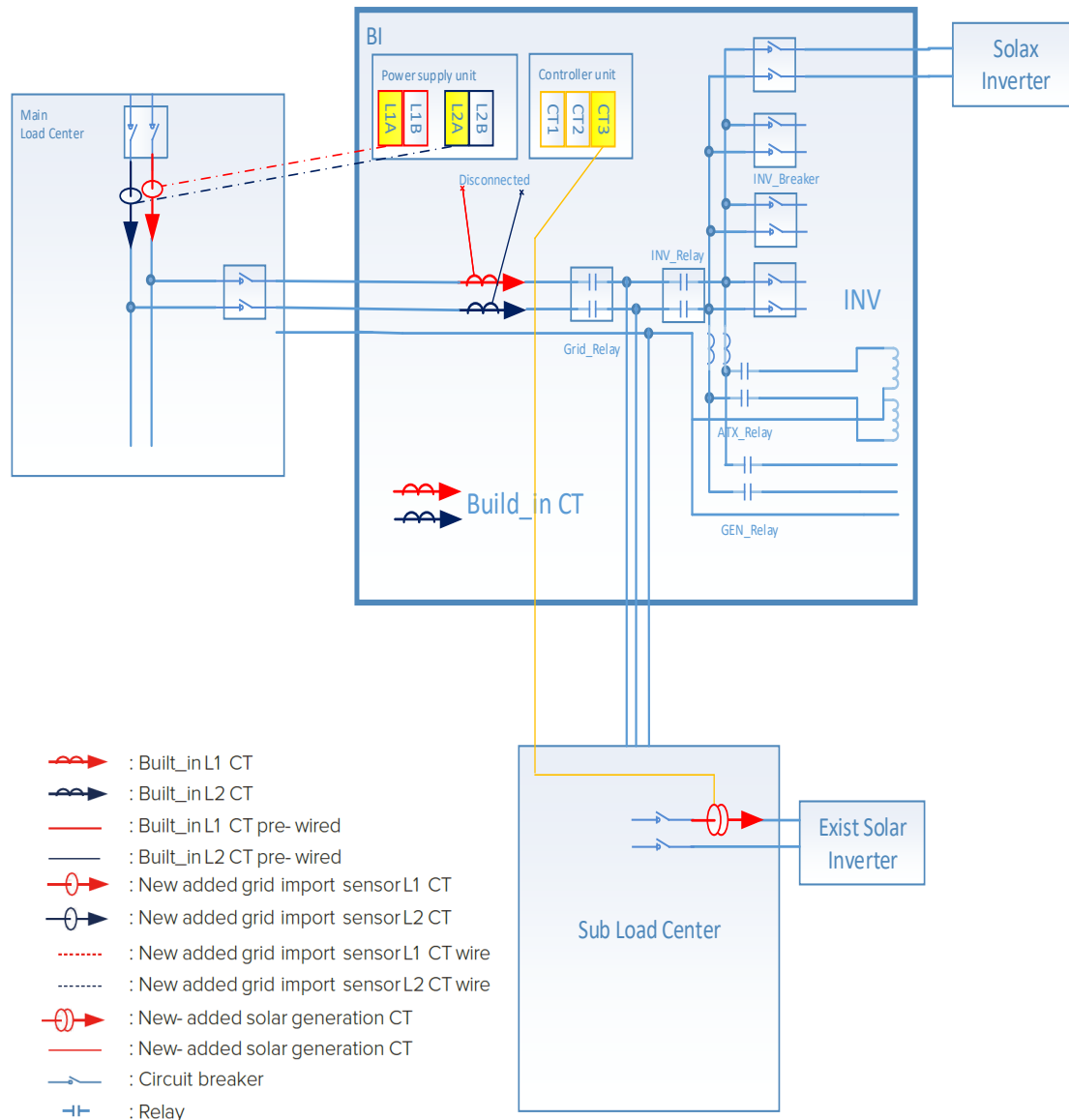


2.4 Partial home backup with third party grid tied solar inverter

Disconnect the pre-installed CT and re-connect the new added CT to the L1A and L2A respectively.

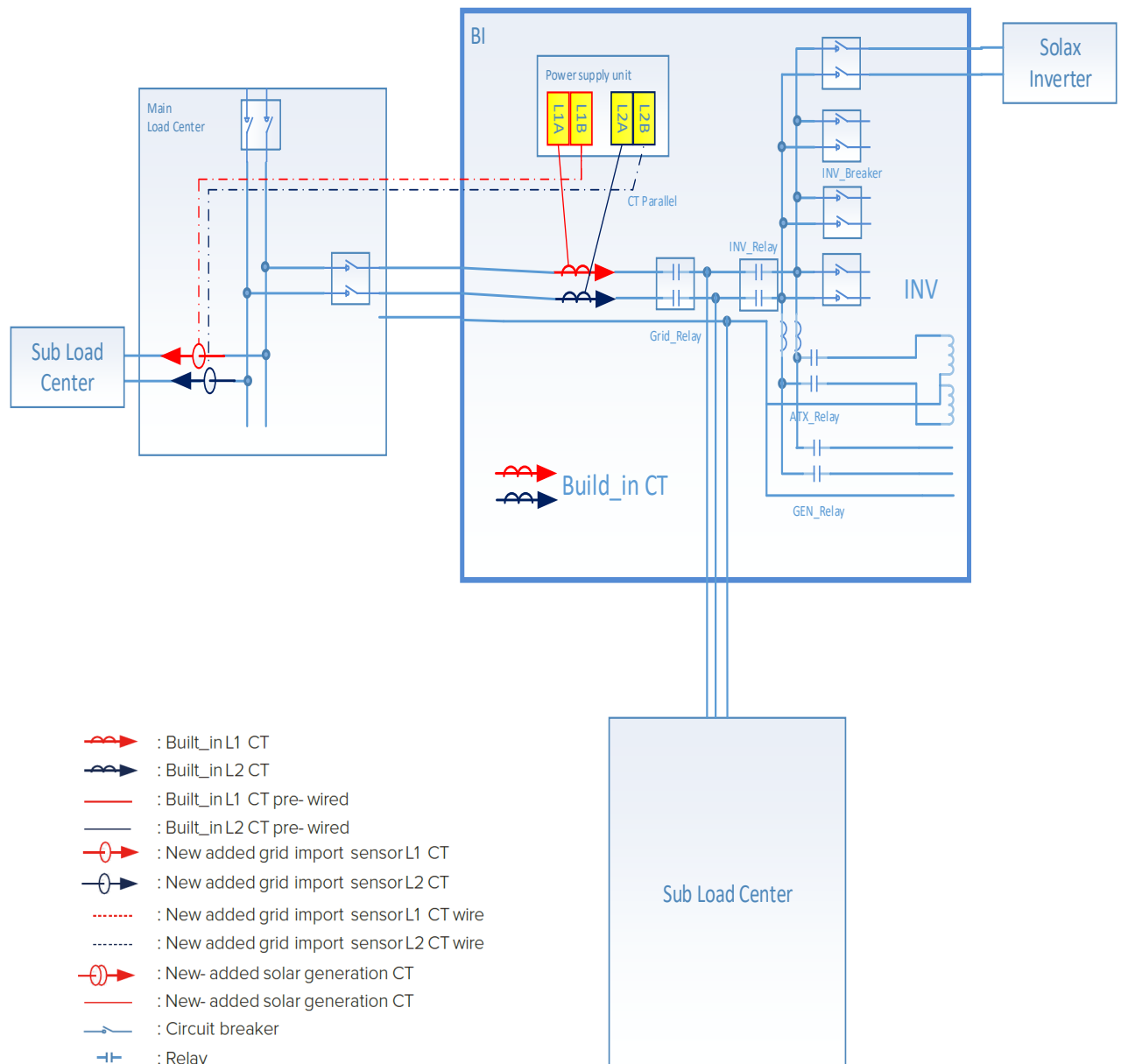
Need two extra CT to monitor the grid input currents.

Need a new extra CT to monitor the current of solar inverter. This new extra CT should be connected to "CT3" connector on the controller unit.



2.5 Partial home back up with different panels

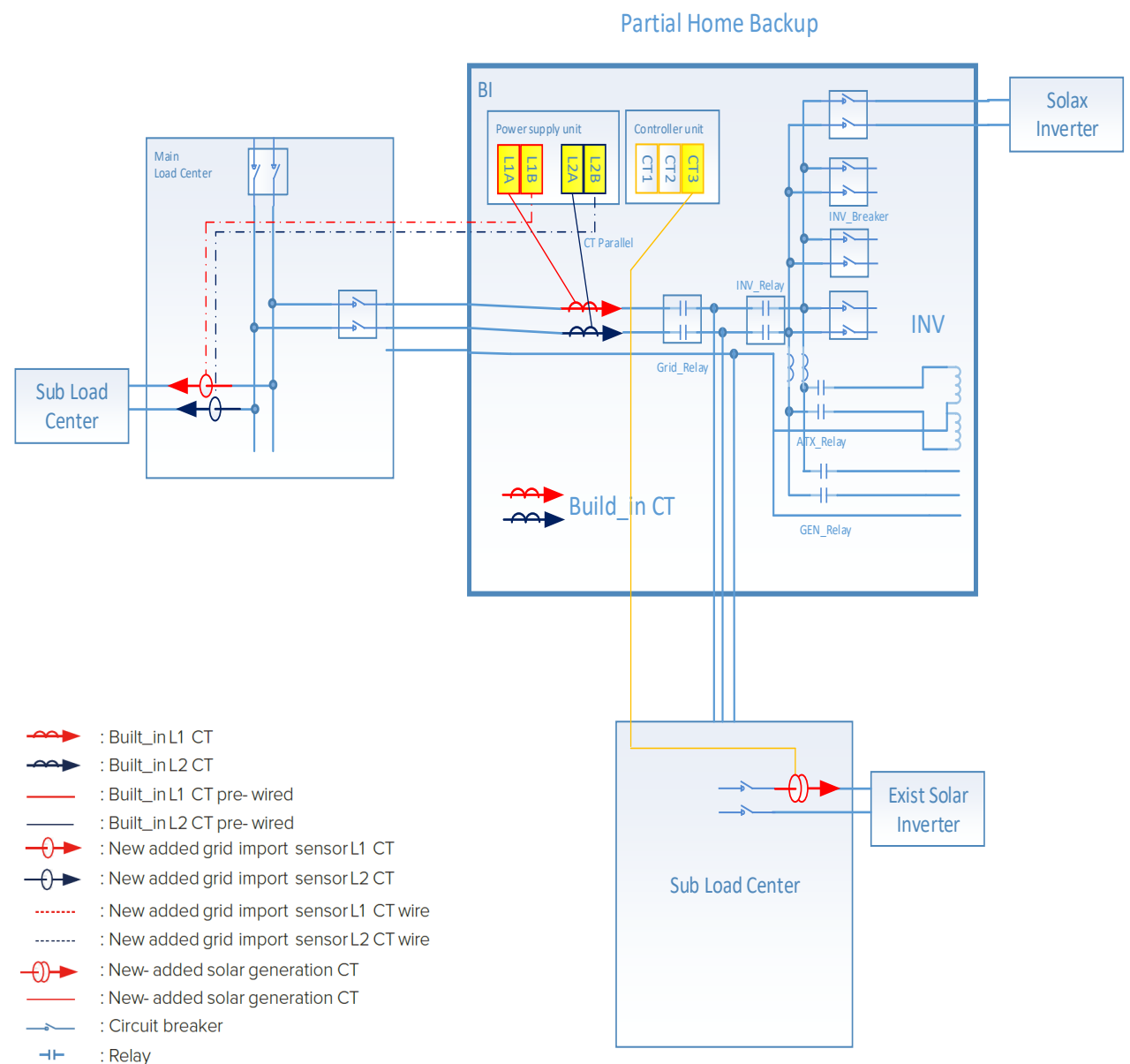
Need two extra CT to monitor the grid input currents.



2.6 Partial home back up with third party grid tied solar inverter, with different panels

Need two extra CT to monitor the grid input currents.

Need a new extra CT to monitor the current of solar inverter. This new extra CT should be connected to "CT3" connector on the controller unit. And the "CT3" clamp must connect to the grid tied solar inverter L1.



3. APP configuration

App configuration function is under development and it will be released in Q1 end of 2023.

4. CT install requirement

- The CTs wires should be routed near the right side of the BI wall internally. The CTs linked to the BI controller board can also be routed at the left upper corner.
- The CTs wires should be tied tightly, and make sure do not contact the bus bar.
- All the CTs directions are from grid to load.
- All the CTs phase should be the same as the voltage. Check the voltage between the grid port of BI L1 voltage and the conductor clamped by the L1 CT, if the voltage is around zero, the CT phase is right.
- All CTs should be clamped at the right location.

5. Tool Requirement

Free tool. The accessories are terminated with the same connectors, just plug.